

Form of Explanation: Explainability vs. Explicability of an Automation Aid under Multitasking Load

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Introduction: TODO

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1. Introduction

People increasingly supervise rather than directly operate automated systems, from aviation flight decks to medical monitoring and driver-assistance aids. In these settings the human is expected to step in when the automation reaches its limits, which only works if the operator understands what the system is doing. The standard procedure to achieve this is to make systems *transparent* and to have them *explain* their behaviour. Yet producing an explanation is not the same as the explanation being understood and acted upon — a distinction we sharpen in Section 2 as *explainability* versus *explicability*. Indeed, work in explainable artificial intelligence (XAI) reports a worrying dissociation: explanations can raise users' confidence and feeling of being informed without improving their ability to detect when the system is wrong (Bansal et al., 2021; Wang & Yin, 2021).

This report asks whether the *linguistic form* of an explanation — its wording, framing, and presentation — changes that dissociation when the information content is held constant and attention is divided across competing tasks. We compare three forms of explanation delivered by an automation aid in a multitasking environment and measure both a subjective response (perceived transparency) and an objective indicator of whether the operator's mental model was actually updated. The remainder of the report is structured as follows. Section 2 reviews the relevant lecture content and literature and defines our central terms. Section 3 states the research gap, question, and hypotheses. Section 4 describes the study design, materials, procedure, and measures. Sections 5–7 report the results, discuss them in relation to prior work, and conclude.

2. Background

2.1 Definitions of terms

We distinguish three related but separate constructs that recur throughout the lecture and the literature.

Transparency is the degree to which a system provides the operator with insight into its inner workings, intentions,

and reasoning process. Frameworks such as the Situation Awareness-based Agent Transparency (SAT) model describe transparency in terms of *levels* of information the agent communicates — what it is doing, why, and what it predicts (Chen et al., 2014; Mercado et al., 2016).

Explainability is the system's act of producing an explanation of what it did and how. It is a property of the system's output.

Explicability is whether that explanation is presented in a way that it is actually understood and processed by the operator. It is a property of the human–system interaction, not of the output alone. Two explanations can be equally explainable (containing the same facts) yet differ in explicability.

2.2 Relation to the lecture

The project is anchored in two lecture topics. The *transparency* block addresses the question of what level of information an aid should communicate to the operator. The *explainability and explicability* block introduces criteria for good explanations — among them *contrastiveness* and *actionability*. These criteria, drawn from the social-science account of explanation (Miller, 2019), motivate our design. Crucially, we deliberately do *not* manipulate selectivity: selecting fewer facts removes information and would be in line with existing research and would not support the research gap we want to target. We therefore hold the information content constant and isolate *contrastiveness* (how the same facts are framed) and *actionability* (an added operator cue).

2.3 Prior empirical work

Transparency studies in human–agent teaming show that giving operators more information about an agent's reasoning and uncertainty can improve trust calibration and decision quality (Mercado et al., 2016; Stowers et al., 2017). These studies vary the *level* of information but do not separate information *content* from *presentation*. In parallel, XAI research on AI-assisted decision-making finds that explanations frequently increase users' agreement and subjective sense of understanding without a matching increase in objective understanding or error detection (Bansal et al., 2021; Wang &

Yin, 2021). Crucially, these dissociation findings come almost entirely from single-task settings, where the user can devote full attention to the explanation.

2.4 Research gap

Two gaps follow. First, prior transparency work does not distinguish the content of an explanation with its linguistic form; it is therefore not known whether linguistic form alone — holding information content constant — affects perceived transparency and the operator’s mental model. Second, the explainability–explicability dissociation has been demonstrated under full attention but not under *multitasking load*, the very condition under which supervisory operators usually work. If a long, status-log-style explanation makes operators feel informed without enabling them to anticipate the system’s next action, they may miss automation errors with far-reaching consequences in domains such as aviation, medical monitoring, and autonomous driving aids.

3. Research Question and Hypotheses

Research questions. Both ask whether the *linguistic form* of an automation aid’s reasoning explanation — holding information content constant under multitasking load — matters, but they target the two different aspects of the explainability–explicability dissociation: the subjective feeling of being informed and the objective mental-model update.

RQ1 (subjective). Does the form of the explanation change the operator’s *subjective transparency* — the feeling of being informed? (*addressed by H1*)

RQ2 (objective). Does the form of the explanation change whether the operator’s *mental model actually updates* — objective explicability? (*addressed by H2*)

We compare three forms of explanation that all communicate the same facts (see Section 4.2.1). The comparison is therefore between *linguistic forms* of explanation, not between the amount of information provided. Hypotheses H1 and H2 are the load-bearing pair: The fact that they predict opposite directions is a key sign that explainability and explicability are not the same. H3 and H4 are supporting hypotheses.

H1 (verbose feels more transparent). Subjective transparency is highest under the verbose form (F1), even though the information content is identical across forms.

H2 (contrastive form improves explicability). Post-block mental-model accuracy is highest under the contrastive forms (F2/F3), even though the information content is identical across forms.

H3 (contrastive form improves miss detection). Because the action-relevant fact is easier to read off a contrastive panel, participants detect a higher proportion of automation-missed events under F2/F3 than under F1.

H4 (saved reading time is redeployed). Participants achieve higher performance on the non-aided tasks (communications and resource management) under F2/F3 than under F1.

The reasoning behind H1 is that a detailed stream of information makes the system *feel* maximally informative. H2–H4 follow from the prediction that contrastive framing and an actionable cue make the same information easier to process under load, freeing attention and directing it to the events that matter.

3.1 Reference to other groups

Most groups built on the lecture constructs: transparency, trust, mental models, workload. The closest to us vary the *wording* of the aid’s messages: **AH Group** (actionable vs. generic), **Group P** (concrete vs. vague warnings), **Group A** (calm vs. neutral tone), team **It-takes-two** (reason-giving vs. status-only), and especially **Group K**, which compares no/vague/clear failure messages with length matching and a comprehension check. Other groups are more distant: they add an explanation rather than reframing it, manipulate the system itself (reliability, automation level, or task load) rather than its message, or treat trust as a personal trait or social signal.

What is unique to us is that we hold the *information content constant* across all three forms, isolating linguistic form rather than “more vs. less” explanation; and we separate *explainability* from *explicability*, predicting them to move in opposite directions (the verbose form feels most transparent, H1, yet the contrastive forms produce the better-updated mental model, H2). Where the majority of our colleagues asks “does clearer explanation raise trust?”, we ask whether an explanation can raise the *feeling* of being informed without improving what the operator actually *knows* — captured by an objective mental-model probe that, across all groups, only we use.

4. Method

4.1 Participants

TODO

4.2 Materials

The study runs on **OpenMATB** (Cegarra et al., 2020), an open-source reimplement of NASA’s Multi-Attribute Task Battery (Comstock & Arnegard, 1992). OpenMATB presents several supervisory subtasks concurrently; participants actively perform all three of them: *system monitoring* (watching gauges and warning lights and responding to out-of-range states), *communications*, and *resource management* (pump and tank management). System monitoring is the only subtask supported by an automation aid; the other two compete for attention and are never aided.

The automation aid operates on the system-monitoring task at approximately 78% reliability. Each block contains

nine events, one every 30s: five *routine* events the aid handles in time, two *near-misses* the aid handles just in time, and two *misses* on which the aid does not act, so that — if the information is perceived and processed — the operator can still handle the event manually (7 of 9 handled appropriately $\approx 78\%$). The aid’s behaviour, reliability, and failure timeline are identical across conditions; only the *linguistic form* of the explanation panel changes.

4.2.1 The three forms of explanation

A panel fires on every event in every form (nine times per block), reporting the same facts: which indicator, whether it was in or out of range, whether the aid acted, and (for near-misses) the time-to-failure. The three forms differ only in their wording (Table 1). The verbose form adds grammatical scaffolding and redundant exact sensor values but introduces no new content with respect to catching a failure. This content matching is the core of the design: the information an operator needs to catch an automation failure is equally present in F1, F2, and F3, so the $F1 \leftrightarrow F2$ contrast isolates *contrastiveness* and the $F2 \leftrightarrow F3$ contrast isolates *actionability*.

Table 1: The three forms of explanation, illustrated on a *miss* (the aid did not act, so the operator must intervene). All forms report the same action-relevant facts and fire on every event; F1 embeds the facts in a verbose routine stream, F2 reframes them contrastively, and F3 adds an explicit operator cue on misses only.

	Description and example panel text
F1	<i>Verbose.</i> Full-sentence status report on every cycle with exact sensor values. E.g. “Cycle 05 (auto-aid) — Scale-1 was 48.0, above the upper bound of 45.0; the automation did not act on this gauge.”
F2	<i>Contrastive.</i> Same facts, tight and contrastive. E.g. “Skipped scale-1 — auto-aid did not act.”
F3	<i>Contrastive + actionable.</i> F2 plus an explicit operator cue on misses. E.g. “Skipped scale-1 — auto-aid did not act. <i>Check it yourself.</i> ”

4.3 Design

The study uses a within-subjects design. Each participant completes all three forms (F1/F2/F3), each paired with one of three different gauge sets (A, B, C). The gauge sets contain different subsets of gauges and lights requiring attention, so that the post-block memory probe cannot be answered from another block’s content. The order of F1/F2/F3 is counterbalanced across participants with a Latin square to mitigate learning and order effects; because blocks are crossed with forms, every form meets every gauge set equally often.

4.4 Procedure

Each session lasts approximately 20 minutes: a 3-minute practice phase, followed by three 5-minute experimental blocks (one per form), followed by a debriefing. After each block questionnaires are used in a fixed order — the objective mental-model probe *first* (to capture the mental model before reflection contaminates it), then subjective transparency, then the single workload item *last*. One final preference questionnaire is completed at the end.

4.4.1 Briefing and debriefing

TODO

4.5 Measures

Each measure maps onto one hypothesis. The questionnaire items are OpenMATB generic scales presented after each block in the fixed order; every slider starts at a neutral position. The two behavioural measures (H3, H4) are read from the continuous platform log.

4.5.1 H1 — Subjective transparency

Three sliders capture the *feeling* of being informed (anchors in brackets):

- “The system told me what it was doing.” (*strongly disagree–strongly agree*)
- “It was hard to keep track of what the automation was doing.” (*strongly disagree–strongly agree, reverse-scored*)
- “The amount of information the system gave me felt about right.” (*strongly disagree–strongly agree*)

TODO: argue why these items and not others (literature)

4.5.2 H2 — Mental-model probe

Three yes/no sliders measure whether the operator’s mental model actually updated. Each item targets one *active* gauge of the block by its role: within each block the aid *handles* one gauge (acts on it, and both near-misses land there) and *misses* the other (it never acts on it). Every correct answer is therefore fixed by what happened in *this* block, not by a constant strategy learned once and reused (anchors in brackets):

- “Did the automation act on [gauge] at any point during this block?” (*no–yes; correct answer Yes for the handled gauge*)
- “Did the automation FAIL to act on [gauge] at some point, leaving you to handle it yourself?” (*no–yes; correct answer Yes for the missed gauge*)
- “Was there a close call (near miss) on [gauge] during this block?” (*no–yes; correct answer Yes for the handled gauge, which carries both near-misses*)

The probed gauges and their roles are counterbalanced across blocks, so the correct-answer pattern (Yes/Yes/No in one block, No/No/Yes in the others) embeds no fixed response heuristic. The responses are scored offline against the block ground truth to create an objective explicability score.

4.5.3 H3 — Detection of automation misses

Read from the platform log and count the number of automation-missed events (two per block) that the participant successfully detects and handles manually. The proportion of detected misses is compared across forms. For caught misses, calculate the reaction time from the moment the aid's panel fires to the moment the participant clicks on the relevant gauge or light.

4.5.4 H4 — Non-aided task performance

H4 tests whether the attention freed by an easier-to-read explanation panel is redeployed to the two tasks the aid never handles: resource management and communications. Both are read directly from the continuous OpenMATB platform log, so no questionnaire is involved. We report two measures per task, chosen to be the standard MATB readouts for that task (Cegarra et al., 2020; Comstock & Arnegard, 1992; Santiago-Espada et al., 2011).

Resource management. Two depleting tanks (A and B) must be held near a target level of 2500 units. We report (i) the **root-mean-square deviation (RMSD)** of the tank level from target, averaged over the two tanks (lower is better), and (ii) the **proportion of time the tanks stayed within tolerance** (± 500 units, i.e. the 2000–3000 band; higher is better). RMSD captures the typical size of the deviation the operator allowed to build up, while time-in-tolerance captures how much of the block was spent in an acceptable state; together they index how well the unaided regulation task was maintained (Comstock & Arnegard, 1992; Santiago-Espada et al., 2011).

Communications. The operator must respond to own-call-sign radio requests by tuning the named radio to the requested frequency, while ignoring distractor call signs. We report (i) **response accuracy**, the proportion of own-call-sign trials answered correctly (higher is better), and (ii) the **response time on hits**, the latency in milliseconds from the end of the call to a validated response (lower is better). Accuracy indexes how reliably the secondary auditory task was serviced, and response time indexes how promptly, which is the more sensitive marker of spare attentional capacity under load (Cegarra et al., 2020).

4.5.5 Workload check

A single mental-effort slider after each block:

- “How mentally demanding or overwhelming did this block feel?” (*very low–very high*)

TODO: argue why this item is necessary and sufficient to check the mechanism of multitasking load.

5. Results

TODO

6. Discussion

TODO

7. Conclusion

TODO

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